

The empirical recurrence for OEIS A234078: recovering a conjecture that is not written down

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Abstract

OEIS A234078 records “Empirical recurrence of order 97”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 625$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 97, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 97 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234078 is “Number of $(n+1) \times (3+1)$ 0..4 arrays with every 2×2 subblock having the sum of the absolute values of all six edge and diagonal differences equal to 10.”. It has offset 1 and begins

2352, 18048, 139512, 1219868, 10952176, 105917092, 1033754752, 10502679784, ...

The entry states:

Empirical recurrence of order 97 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:09:25, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 625$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 625$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1250$ exact terms of the model returns order 97 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{97} c_i a(n-i),$$

c_1	0	c_{26}	2410941839715772046870	c_{51}	22497716934854426514815586790
c_2	455	c_{27}	6693825461716724940654	c_{52}	−2189230669320006427317348149
c_3	200	c_{28}	−21828209835021144312368	c_{53}	−3953803371926220169776573671
c_4	−93330	c_{29}	−63665373280964415134736	c_{54}	8672837907111884007682415660
c_5	−74614	c_{30}	167792458511720135223146	c_{55}	61045347367193719508336517706
c_6	11529141	c_{31}	515013330776263967043812	c_{56}	2627581027801650246688110241
c_7	12689680	c_{32}	−1100047467091352413512045	c_{57}	−8277505510689609780609632789
c_8	−966312208	c_{33}	−3564849673032258202275428	c_{58}	−8195357914595637728639595873
c_9	−1311606914	c_{34}	6172911843682779156801196	c_{59}	98497595902026944797309813719
c_{10}	58670732962	c_{35}	21224843661046729839472520	c_{60}	14488230734682326543455840113
c_{11}	92664564072	c_{36}	−29725900600458619844774537	c_{61}	−10273461040651166576408926788
c_{12}	−2689595511716	c_{37}	−109192012571135695800585188	c_{62}	−1933824495627561554333572306
c_{13}	−4771834526564	c_{38}	123024352699758069559994885	c_{63}	93770265483372253244147879593
c_{14}	95786809974547	c_{39}	487262054933407365968868760	c_{64}	20931073398689098417972783185
c_{15}	186537265612734	c_{40}	−437639617512442345342287792	c_{65}	−7474031617494870742785810565
c_{16}	−2706202845320539	c_{41}	−1892282849490229488246618712	c_{66}	−1888620618686693117082742141
c_{17}	−5693603941547564	c_{42}	1335937356864053144489276779	c_{67}	51884945948234958683146159327
c_{18}	61635984759496516	c_{43}	6412861127840925621766672332	c_{68}	14382992382670558045973607348
c_{19}	138550693345071828	c_{44}	−3484225872662711567041334864	c_{69}	−3126953296899026158856972261
c_{20}	−1146326386083770767	c_{45}	−19007772286591075525163200836	c_{70}	−9294472407622460509107719247
c_{21}	−2732031867110823006	c_{46}	7695091758987761760752588417	c_{71}	16296454955294677357922017117
c_{22}	17594535265986666089	c_{47}	49361648881448694236067860898	c_{72}	5103535825828306663400139801
c_{23}	44231608808900499698	c_{48}	−14137822249435225080892775872	c_{73}	−7309711794820313014171778094
c_{24}	−224857724139343494215	c_{49}	−112459859247753049144443489262	c_{74}	−2378085090617625648895286056
c_{25}	−594446194759113715004	c_{50}	20779321353871249164668476637	c_{75}	28057896227611179447247341023

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 97, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $97 + 4$ terms removed returns order 97 as well, so $\deg q_{\min} = 97 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $97 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 97 that this sequence satisfies — and that family turns out to have exactly one member.

References

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